

T1 Contributions Analysis Report

DiscountMate · Data Analytics / Machine Learning Team · Trimester 1 → Trimester 2 handover reference

Author: Robin Le – Sr Cross lead

1. Overview

This report reorganises T1 contributions from the DataBytes DiscountMate handover, grouping them by the six main features planned in the T1 roadmap, then aligning those features to the three T2 objectives set for this trimester. It is intended as a shared reference for the T2 DA/ML team: what was actually built, by whom, and where the gaps are. Where a contributor used labels that diverged from the roadmap, work has been remapped to the closest roadmap feature — sub-task IDs are preserved as recorded in the handover.

1.1 T1 features and implementation status

Feature ID	Feature name	Status	Primary contributor(s)
DA-01	Insight Reporting & Visualization	Live (customer-facing)	Adam Creek, Steve Nguyen
DA-02	Retail Discount Leadership & Follow-On Pricing Pattern Analysis	Delivered (analysis + DAX)	Ahmed Khalaf, Radia Zayeen, Steve Nguyen
DA-03	Product Deduplication & Matching	Delivered	Shivansh Aneja, Ahmed Khalaf
DA-04	Website Analytics (GA4)	Live (production)	Adam Creek
DA-05	Persona to Product Mapping	Design-stage only (exploratory framework)	Adam Creek (exploratory only)
DA-06	Price Volatility and Elasticity	Delivered (framework + dashboard)	Saksham Sachdeva
DA-07 *	Discount Classification & Prediction (off-roadmap)	Delivered (V1, binary)	Adam Creek

* [DA-07] was not in the original T1 roadmap. Adam Creek introduced it during the trimester as a standalone discount classifier feature. It has been included here because it materially feeds T2 Objective 3.

1.2 T2 objectives and T1 feature mapping

The three T2 objectives inherit and extend the T1 work as follows:

T2 Objective	T1 features feeding into it
Objective 1 — GA4: behavioural data / interaction metrics + persona-mapping	[DA-04] Website Analytics · [DA-05] Persona to Product Mapping · DL cross-team (upstream input reference): [DL-02] Receipt OCR Pipeline (Sriniketh Reddy Dumbala)
Objective 2 — PowerBI dashboard: extend to stakeholder-specific views + integrate directly into web's product comparison page	[DA-01] Insight Reporting & Visualization · [DA-02] Retail Discount Leadership (analytics feeding dashboards) · [DA-03] Product Deduplication (feeder for like-for-like comparison)
Objective 3 — Discount classifier / pricing intelligence: advance discount analysis model to production-ready	[DA-06] Price Volatility & Elasticity · [DA-07] Discount Classification (Adam) · DL cross-team: [ML-06] True Value Promotion (Sharon Roy) · [DL-03] Forecasting baselines (Shaeera Islam, Daniel Eikelis)

Objective 3 draws contributions from both the T1 DA team and the T1 DL team. This overlap reflects that T2 has consolidated ML work under the DA/ML team, and confirms the scope collision worth resolving with the AI/DL team before committing T2 deliverables.

2. Detailed Contributions Summary

2.1 Objective 1 — GA4 behavioural data & persona-mapping

GA4 tracking was successfully deployed to production in T1 but stayed at the baseline event level (pageviews / clicks). Persona work remained at design stage with no code implementation, confirmed by a GitHub audit of the T1 repository — although an exploratory personas framework by Adam Creek was found in the repository (not declared in his handover contributions).

DA team contributions

[DA-04] Website Analytics

Owner(s): Adam Creek (DA Stream Lead)

- **DA-04-T1** Research suitable analytics platforms (GA4 selected)
- **DA-04-T2** Modify code snippet for GA4 in React build file
- **DA-04-T3** Conduct local test on GA implementation
- **DA-04-T4** Implement analytics code to PROD env
- **DA-04-T5** Test tags to ensure beacons fire to correct report suite
- **DA-04-T6** Document final setup
- **DA-04-T7** Develop reports for key stakeholder groups

Artifact: Google Analytics Implementation Guide.pdf; GA property live at discountmate-prod.web.app (Measurement ID G-KV1PBPHM30).

Status: LIVE. Reports at baseline event level; no behavioural analysis layer or custom event schema built yet.

[DA-05] Persona to Product Mapping

Owner(s): Adam Creek — exploratory framework only. All roadmap sub-tasks unimplemented.

- **DA-05-T1** Review and finalise previous Persona Construct — not started (see exploratory framework note below)
- **DA-05-T2** Ensure Persona pre-processing on new data — not started
- **DA-05-T3** Develop Python script to map personas to products — not started
- **DA-05-T4** Develop Power BI persona-to-product visualization — not started
- **DA-05-T5** Advise / coordinate with Frontend on web application treatment — not started

Artifact: DiscountMate Personas Draft (Adam Creek) — exploratory framework in the T1 GitHub repository, not declared in Adam's handover contributions. Defines five proposed personas (Budget-Conscious Shopper, Health Enthusiast, Family Planner, Convenience Seeker, Premium Shopper), a sign-up decision-tree for persona allocation, and a rule-based keyword tagging table for product-to-persona mapping. Useful as a review starting point for T2's shift to behaviour-derived segmentation.

Status: DESIGN-STAGE ONLY. The largest inherited gap; primary work item under T2 Objective 1.

DL team cross-contribution (potential upstream input reference)

[DL-02] Receipt OCR Pipeline

Owner(s): Sriniketh Reddy Dumbala — noted for reference only;

- **DL-02-T1** Implement receipt image preprocessing and validation using OpenCV
- **DL-02-T2** Implement OCR text extraction pipeline for supermarket receipts (Tesseract)
- **DL-02-T3** Integrate OCR processing workflow with backend API endpoints
- **DL-02-T4** Implement receipt parsing and initial product-matching functionality
- **DL-02-T5** Implement authenticated receipt-history storage into MongoDB user profiles

- **DL-02-T6** Perform local end-to-end backend testing and OCR workflow validation

Artifact: PR #259: https://github.com/DataBytes-Organisation/DiscountMate_new/pull/259 | PR #273: https://github.com/DataBytes-Organisation/DiscountMate_new/pull/273

Status: DELIVERED. Potential upstream input to Objective 1 — receipt purchase history is referenced in the T2 plan as a data source for behaviour-derived persona segmentation. The broader T1 DL OCR effort (Catalogue OCR by Kevin Kearney and Steven Le, under [DL-01]) is not directly relevant to Objective 1, as its output is product intelligence rather than user behavioural signals.

2.2 Objective 2 — PowerBI dashboard: stakeholder views & product-page integration

The customer-facing Power BI dashboard was successfully deployed to production, with the underlying analytical layer (discount leadership analysis, cross-retailer lag models, private/house-brand segmentation, product deduplication) all delivered. Stakeholder-specific views and product-page embedding were not attempted in T1 — these are the extension vectors for T2.

[DA-01] Insight Reporting & Visualization (Power BI dashboard + data foundation)

Owner(s): Adam Creek, Steve Nguyen (Jr Lead)

- **DA-01-T1**
Adam: Re-establish Power BI connection to MongoDB data
Steve: Establish end-to-end ETL pipeline via SharePoint for ALDI, Coles, IGA, Woolworths
- **DA-01-T2**
Adam: Refresh data from latest pipeline outputs via SQL Endpoint
Steve: Author Power Query M-scripts to standardise retailer headers into a unified schema
- **DA-01-T3**
Adam: Build dashboards for pricing / trend analysis
Steve: Data auditing to isolate extreme marketplace pricing skews (> \$100)
- **DA-01-T4**
Adam: Investigate and test nightly data refresh
Steve: Engineer server-side price-ceiling 'Hard Filter' for like-for-like grocery comparison
- **DA-01-T5**
Adam: Update documentation
- **DA-01-T6**
Adam: Handover ownership to junior team member
- **DA-03-T1 to DA-03-T4 (Steve's labels — dashboard production deployment phase)**
DA-03-T1: Design front-end visual canvas layers: price gap indicators, avg price bars, KPI cards
DA-03-T2: Synchronise reporting fields with MongoDB/Postgres backend
DA-03-T3: Configure cloud workspace permissions + secure embedded iframe web link
DA-03-T4: Coordinate with Web Dev to embed dashboard into production portal

Artifact: Live dashboard: <https://discountmate-prod.web.app/product-dashboard> | Power BI file: DataBytes_main.pbix | PowerBI-Mongo DB Connection Guide.pdf

Status: LIVE. Steve's [DA-01] and [DA-03] sub-task IDs are his own labels, remapped here to the roadmap's DA-01 since both scopes deliver the dashboard end-to-end (data foundation + BI UI + production deployment).

[DA-02] Retail Discount Leadership & Follow-On Pricing Pattern Analysis

Owner(s): Ahmed Khalaf, Radia Zayeen, Steve Nguyen

- **DA-02-T1** Dataset cleaned and structured — cross-retailer master dataset, size normalisation, nested folder handling
- **DA-02-T2** 2–3 product categories analysed with justification — branded vs private-label; curated matching rules for milk, cola, corn chips
- **DA-02-T3** Discount events correctly identified — price-drop events grouped into waves
- **DA-02-T4** Price leader identified for each discount event — first-discounting retailer flagged
- **DA-02-T5** Follower retailers and lag time (days) calculated — Leader-Follower Analysis complete

- **DA-02-T6** (Steve — DAX layer) Basket calculation engine using SUMX/VALUES iterator logic for a 10-item 'Hero Basket'
- **DA-02-T7** (Steve — DAX layer) Automated time-intelligence chain for Week-over-Week price shift %
- **DA-02-T8** (Steve — DAX layer) Predictive cross-retailer temporal lag model using DATEDIFF
- **DA-02-T9** (Steve — DAX layer) Text-matching classification framework to segment private/house brands from national brands

Artifact: Leader/follower tables, price series, markdown reports (Radia). DAX semantic layer embedded in DataBytes_main.pbix (Steve). No standalone script link provided in handover for Ahmed/Radia's Python analysis.

Status: DELIVERED. Analytical layer complete; findings surface in the live dashboard via the Power BI DAX model. Note: DA-02-T6 to T9 are Steve's DAX-layer items originally labelled [DA-02-T1], [DA-02-T2], [DA-02-T5], [DA-02-T6] — renumbered here to avoid collision with the roadmap sub-tasks.

[DA-03] Product Deduplication & Matching

Owner(s): Shivansh Aneja, Ahmed Khalaf

- **DA-03-T1** Clean and standardise product names — lowercase, strip special chars/punctuation, remove store-specific words and unit values (e.g. '500g', '1L')
- **DA-03-T2** Extract brand and key tokens from product titles
- **DA-03-T3** Implement token-based matching logic across Coles, Woolworths, ALDI, IGA using match keys and similarity checks
- **DA-03-T4** Validate and review matched products; add confidence levels and validation rules

Artifact: No standalone artifact link in the handover; matching scripts live in the T1 repository.

Status: DELIVERED. Foundational for cross-retailer like-for-like comparison; both dashboard and pricing intelligence work depend on it.

2.3 Objective 3 — Discount classifier & pricing intelligence (production-ready)

T1 shipped two V1 discount-scoring features from separate teams: Adam Creek's binary real-vs-fake classifier from the DA team, and Sharon Roy's category-aware True Value Promotion scoring from the DL team. Forecasting baselines were built by the DL team but not deployed. This is the objective with the most cross-team overlap — therefore T2 planning will focus on developing one model (**True Value Promotion model**) only.

DA team contributions

[DA-06] Product Price Volatility & Elasticity

Owner(s): Saksham Sachdeva

- **DA-06-T1** Merging datasets and data cleaning
- **DA-06-T2** Research approach to pricing analysis
- **DA-06-T3** Develop Python script to implement analysis (price volatility, coefficient of variation / Price CV framework)
- **DA-06-T4** Develop visualisation / report (Python + Power BI dashboards with drill-down by product and retailer)
- **DA-06-T5** Advise and coordinate with Frontend Team on web application treatment

Artifact: Framework built in Python + Power BI. No standalone artifact link provided in handover.

Status: DELIVERED. Provides Price CV metrics for evaluating retailer pricing behaviour and promotional impact; feeds into T2 depth-forecasting work by AI/ DL team.

[DA-07] Product Discount Classification & Prediction Model (off-roadmap)

Owner(s): Adam Creek (DA Stream Lead)

- **DA-07-T1** Select category and combine weekly data
- **DA-07-T2** Develop Python script to create baseline price over time series by product
- **DA-07-T3** Compare product price to baseline and classify discount (real vs fake)
- **DA-07-T4** Use classification to predict next week's price

Artifact: Price_Discount_Classification_v3.py

Status: DELIVERED V1 (binary classification). Under the T2 direction, this work transitions to AI/DL as the base for their discount-depth forecasting extension.

DL team cross-contribution

[ML-06] True Value Promotion (DL team feature, complete V1) – main reference for T2 development

Owner(s): Sharon Roy (DL Team Lead)

- **ML-06-T1** Review scraped retailer datasets, select reliable sources for V1
- **ML-06-T2** Harmonise Coles, Woolworths, IGA pricing/promotion data into a master dataset
- **ML-06-T3** Clean and standardise pricing / promotion / category / timestamp fields
- **ML-06-T4** Engineer measurable savings, discount %, and promotion-value indicators
- **ML-06-T5** Build category-aware promotion benchmarking and scoring logic
- **ML-06-T6** Implement customer-facing classifications: OK Discount / Good Discount / Excellent Discount / DiscountMate Recommends
- **ML-06-T7** Validate and refine scoring logic for balanced customer-facing results
- **ML-06-T8** Prepare documentation, result datasets, and future-improvement notes

Artifact: PR #275: https://github.com/DataBytes-Organisation/DiscountMate_new/pull/275 | True Value Promotion Feature Documentation.pdf (Sharon Roy, T1 2026).

Status: SHIPPED V1. Under the T2 direction, this becomes the primary DA/ML pricing intelligence asset — advanced from V1 to V2 with production-readiness as the goal.

[DL-03] Price Forecasting Baselines (deep learning + statistical models)

Owner(s): Shaeera Islam, Daniel Eikelis

- **DE-03-T2** Clean and standardise time-series price data
- **DL-03-T3** Analyse pricing trends and discount cycles
- **DL-03-T4** Engineer features for forecasting models (Prophet)
- **DL-03-T5** Develop baseline forecasting models — Shaeera: LSTM / XGBoost / GRU · Daniel: Prophet / XGBoost
- **DL-03-T6** Evaluate model performance and refine outputs
- **DL-03-T7** Clean and preprocess for time-series data
- **DL-03-T8** Daniel additionally pivoted to Markov / Bayesian model research due to limited historical data availability

Artifact: PR #288 (Shaeera): https://github.com/DataBytes-Organisation/DiscountMate_new/pull/288

Status: BUILT but NOT DEPLOYED. Under the T2 direction, this work stays with AI/DL as the base for their discount-depth forecasting extension. Documented pipeline gaps to close: GCP repointing, automated retraining, coverage beyond current limited product set.

3. Conclusion

3.1 Implementation status by T2 objective

T2 Objective	Component	Status
Obj 1 — GA4 + persona	GA4 tracking (baseline events)	LIVE (production)
	Behavioural event schema / analysis layer	Not started → T2
	Persona construct (rule-based, from prior tri; exploratory framework by Adam)	Design-stage only → T2
	Persona → product Python mapping	Not started → T2
	Persona Power BI visualisation	Not started → T2

T2 Objective	Component	Status
Obj 2 — Power BI	Customer-facing dashboard (retailer/pricing view)	LIVE (production, embedded)
	Underlying DAX analytics (WoW, cross-retailer lag, private brand seg.)	Delivered
	Stakeholder-specific views	Not started → T2
	Product-page embedding on comparison page	Not started → T2
Obj 3 — Pricing intelligence	Discount classifier V1 (binary, real vs fake)	Shipped (Adam)
	True Value promotion scoring V1 (category-aware)	Shipped (Sharon) → T2
	Price volatility & elasticity framework	Delivered (Saksham)
	Forecasting baselines (LSTM / XGBoost / GRU / Prophet)	Built, not deployed
	Discount depth prediction ('how much cheaper')	Not started → AI/ DL team
	Pipeline production-readiness (GCP, retraining, coverage)	Gaps documented, not closed

3.2 Key T2 actions to extend T1 contributions

Actions below are drawn from the T2 Company Report (*DataBytes Task 2.1P T2 2026*) and framed as continuation vectors on the T1 baseline. Objective 3 reflects the T2 scope split negotiated between DA/ML and AI/DL: DA/ML focuses on advancing Sharon Roy's True Value Promotion scoring model; discount depth forecasting transitions to AI/DL.

Objective 1 — Behavioural Analytics & Persona Segmentation

- Extend GA4 tracking beyond pageview / click-events to capture richer behavioural signals (search depth, revisit frequency, discount interaction, retailer preferences).
- Shift personas from rule-based to behaviour-derived segmentation, using GA4 events, grocery list activity, and (where available) receipt purchase history.
- Engineer behavioural features (category affinity, revisit frequency, price sensitivity, retailer preferences) and apply clustering (e.g. K-means) to form data-driven segments.
- Generate behavioural insight reports for the product and recommendation teams.
- Define persona → product mapping and recommender interface in collaboration with AI/DL, specifying how segments feed personalised suggestions and Smart Swaps.
- Framework-first approach: build segmentation logic and interface now; validate as real user volume grows.

Objective 2 — Insight Surfacing (Power BI)

- Extend Power BI dashboards into stakeholder-specific views beyond the current customer-facing dashboard.
- Prototype product-page insight embedding with the Web Dev team to surface pricing and trend insights directly on comparison pages (*currently: dashboard takes place in a separate Product Dashboard page*)

Objective 3 — Pricing Intelligence (revised direction)

- **T2 aim:** DA/ML owns evolution of the True Value Promotion scoring model; AI/DL owns discount depth forecasting.
- **Stabilise and harden V1:** Refactor the V1 notebook into a structured Python module; close Sharon's documented data gaps (re-include Aldi once DE's silver layer standardises promotion fields; improve Woolworths category coverage; refine 'Other' category); add unit-price fairness checks.
- **Evolve V1 → V2 (contextual scoring):** Introduce a historical price context layer to reflect whether a discount is genuinely low for the SKU historically; add a promotion authenticity signal (recurring vs event-driven markdowns);

replace static label thresholds (25 / 45 / 70) with distribution-derived thresholds; add a confidence layer to each scored output.

- **Cross-workstream integration:** Define a Web-Dev-consumable output schema (product ID, label, score, confidence, reason) that feeds Objective 2's product-page embedding.
- **Production-readiness:** Repoint scoring inputs from local files to GCP / silver layer; automate scoring refresh on silver-layer updates; expose scored output via API endpoint; build an offline evaluation harness with team-labeled 'gold' examples.
- **Potential extension — persona-aware scoring:** If Objective 1 progresses meaningfully, layer persona-aware variants on the core score (e.g. 'Budget-Conscious' customer weights discount %, 'Premium' customer weights specialty rarity). High-leverage cross-workstream bet, since only DA/ML owns both surfaces — but keep it optional until persona segmentation is validated.

Enabling workstream — Data Foundation for Modelling: the T2 plan also flags the need to define ML-ready schema requirements (store identifiers, discount metadata, category consistency), implement validation checks, and confirm behavioural and purchase data flows cleanly through the pipeline. This is not a headline objective but underpins all three.

3.3 Summary

T1 shipped a live GA4 property, a customer-facing Power BI dashboard, a rich underlying analytical layer (discount leadership, cross-retailer lag, price volatility), and two independent V1 discount-scoring features from separate teams (Adam's binary classifier, Sharon's category-aware True Value scoring). Forecasting baselines exist but are not deployed. Persona work remains design-stage only, though Adam's exploratory personas framework in the T1 repository provides a review starting point. The T2 team inherits a solid data-and-dashboard foundation, a well-documented True Value Promotion scoring model to be further advanced, and a persona workstream that needs to be built from scratch on behavioural rather than rule-based signals.